

Welfare and Efficiency

1. For the market of baked beans, the demand function is $P=32-8Q$ and supply function is $P=12+2Q$.

- Find the equilibrium price and quantity in this market.
- Plot the demand and supply graph with proper labels. Denote the consumer surplus and producer surplus in your graph.
- Calculate the total surplus.
- Explain why a market operates efficiently at the equilibrium point.

Consumer surplus is the difference between the maximum price you are willing to pay and the actual price you pay.

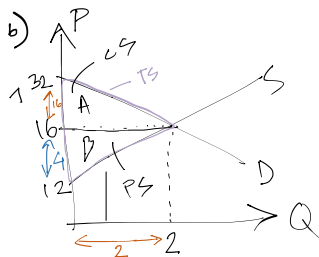
Producer surplus is the difference between the lowest price a producer will sell at and the actual price.

Market interventions such as Price Floors and Price Ceilings create inefficiencies in the market leading to Deadweight Loss (DWL).

a) $D: P = 32 - 8Q$ $S: P = 12 + 2Q$
At equilibrium, $D = S$

$$32 - 8Q = 12 + 2Q$$

$$\therefore Q = 2 \quad \therefore P = 16$$



f.g exact values
Q in both D & S

$$D: P = 32 - 8Q \quad S: P = 12 + 2Q$$

$$\therefore P = 32 \quad \therefore P = 12$$

the maximum price the minimum price

A = consumer surplus B = producer surplus

c)

Area of triangle = $\frac{1}{2} \times B \times H$ $\frac{1}{2} \times 4 \times 2$

$$CS = \frac{1}{2} \times B \times H$$

$$= \frac{1}{2} \times 2 \times 16$$

$$= 16$$

$$PS = \frac{1}{2} \times B \times H$$

$$= \frac{1}{2} \times 2 \times 4$$

$$= 4$$

$$TS = CS + PS = 16 + 4 = 20$$

alternate way

$$\frac{1}{2} \times B \times H$$

$$\Rightarrow \frac{1}{2} \times 2 \times 20 \Rightarrow 20$$

3. Suppose for the market of green tea the demand function is $Q_d = 11000 - 1000P$ and supply function is $Q_s = 500P - 1000$. If a price ceiling of \$4 is imposed on green tea find-

- Equilibrium price and quantity.
- Producer Surplus and Consumer Surplus before the imposition of price ceiling.
- Producer Surplus and Consumer Surplus after the imposition of price ceiling.
- Deadweight loss from the price ceiling.

$$D: Q_d = 11000 - 1000P \quad S: Q_s = 500P - 1000$$

At equilibrium, $D = S$ i.e. $Q_d = Q_s$

$$11000 - 1000P = 500P - 1000$$

$$\Rightarrow 12000 = 1500P \quad \therefore P = \$8 \quad \therefore Q = 500(8) - 1000 = 3000$$

equilibrium price = \$8
equilibrium quantity = 3000

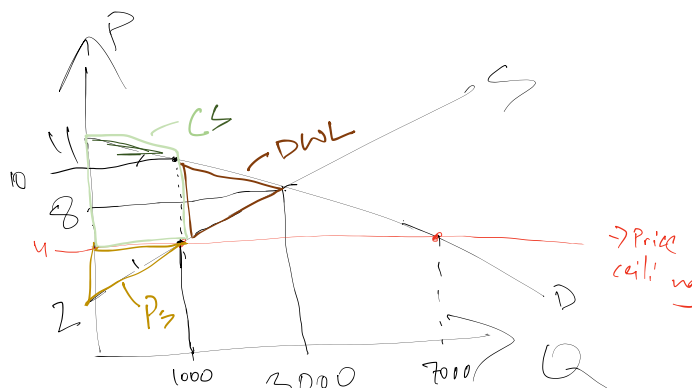
b) At maximum price, $Q = 0$ in D eqn.

$$0 = 11000 - 1000P$$

$$\therefore P = \$11 \text{ (max)}$$

At minimum price, $Q = 0$ in S eqn.

$$0 = 500P - 1000 \quad \therefore P = \$2 \text{ (min)}$$



Before price ceiling

$$CS = \frac{1}{2} \times B \times H = \frac{1}{2} \times 8 \times 3000 = \$12000$$

$$PS = \frac{1}{2} \times B \times H = \frac{1}{2} \times 6 \times 3000 = \$9000$$

$$TS = CS + PS = \$21000$$

c) After price ceiling of \$4, we are finding the D at \$4 & S at \$4

$$D: Q_d = 11000 - 1000(4) = 7000$$

$$S: Q_s = 500(4) - 1000 = 1000$$

Obviously, we cannot operate at Q_D level because suppliers will only sell Q_S level.
(7000) (1000)

Q_S will become new market quantity

$\therefore$ new $Q = 1000$

Finding the value of P in D eqn. when $Q = 1000$

$$1000 = 11000 - 1000P$$

$$\Rightarrow 1000P = 10000 \quad \therefore P = 10$$

$$PS = \frac{1}{2} \times B \times H = \frac{1}{2} \times 2 \times 1000 = \$1000$$

$$CS = \frac{1}{2} \times (a+b) \times H = \frac{1}{2} \times (7+6) \times 1000 = \$6500$$

$$DWL = \frac{1}{2} \times 6 \times 2000 = \$6000$$

Trapezium/Trapezoid = $\frac{1}{2} \times (a+b) \times H$ (2 sides parallel / 2 sides not parallel)

$$TS \text{ (before price ceiling)} = TS \text{ (after price ceiling)} + DWL$$

$$= CS + PS + DWL = 1000 + 6500 + 6000 = \$13500$$

2. Suppose the demand function is $P=200-2Q$ and supply function is $P=50+Q$. If the government introduces a price floor of $P=120$, find-

- Equilibrium price and quantity.
- Producer Surplus and Consumer Surplus before the imposition of price floor.
- Producer Surplus and Consumer Surplus after the imposition of price floor.
- Deadweight loss from the price floor.

a) $D=S$
 $\Rightarrow 200 - 2Q = 50 + Q$
 $\therefore Q = 50 \quad \therefore P = 100$

b) max price, $Q=0$ at Demand.
 $\Rightarrow P = 200 - 2(0) \therefore P = 200$

min price, $Q=0$ at Supply
 $\Rightarrow P = 50 + 0 \therefore P = 50$

Before price floor

$CS = \frac{1}{2} \times B \times H = \frac{1}{2} \times (200 - 100) \times 50 = \2500
 $PS = \frac{1}{2} \times B \times H = \frac{1}{2} \times (100 - 50) \times 50 = \1250
 $TS \text{ (before price floor)} = CS + PS = \3750

c) After price floor of \$120, finding quantity in D & S eqn.

D: $120 = 200 - 2Q \Rightarrow Q = 40$

S: $120 = 50 + Q \Rightarrow Q = 70$

Initially, we need to operate at Q_S level because consumers will only buy Q_D level
 (40)

$\therefore Q_D$ will become new market quantity

$\therefore Q = 40$

We know $P = \$120$ when $Q = 40$ in D eqn.

So, P in S eqn when $Q = 40$

$P = 50 + 40 = \$90$

$CS = \frac{1}{2} \times B \times H = \frac{1}{2} \times (200 - 120) \times 40 = \1600

$PS = \frac{1}{2} \times (a+b) \times H = \frac{1}{2} \times [(120 - 50) + (120 - 90)] \times 40 = \2000

$DWL = \frac{1}{2} \times B \times H = \frac{1}{2} \times (120 - 90) \times (50 - 40) = \150

$TS \text{ (before price floor)} = TS \text{ (after price floor)} + DWL$
 $= CS + PS + DWL = \$3750$

